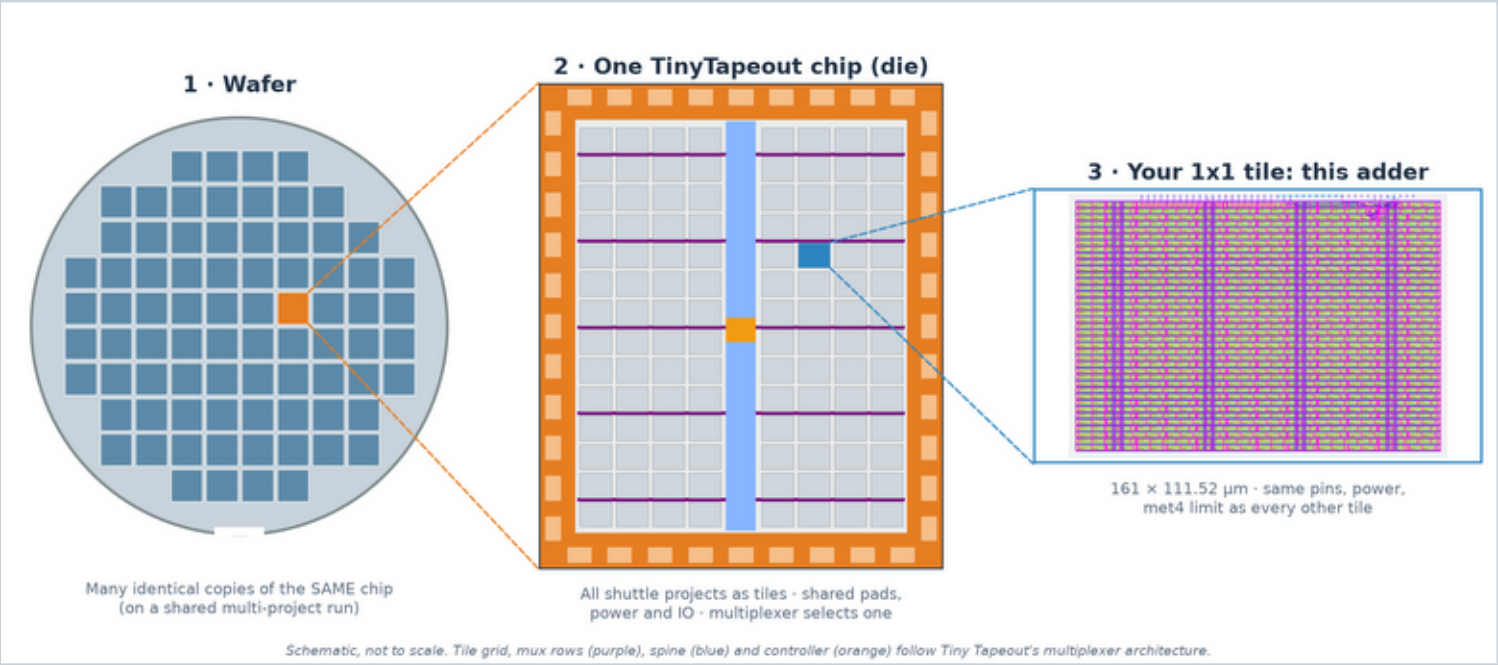
Two-bit unsigned adder

From SystemVerilog to a TinyTapeout SKY130 layout

Open-source flow: Verilator · Yosys · OpenROAD · OpenSTA · Magic · Netgen · KLayout

November 2026

From wafer to your tile



KEY POINTS

- Tiny Tapeout = many designers share ONE chip design, so a real tapeout costs a fraction of a full chip
- The wafer is not split between users: it carries many identical copies of that one chip
- Each chip (die) contains every project of the shuttle as tiles; a 1x1 tile is 161 × 111.52 µm
- Up to 512 designs per chip; tiles sit in rows next to mux units on a central spine
- Rough estimate: die ≈ 3.6 × 5.2 mm (≈ 18.6 mm²), 200 mm wafer → ~1,400–1,600 dice if the wafer held only TT chips; a shared run carries fewer
- **Same pads, power and multiplexer for everyone → identical rules for every tile**

Inside the chip, and what you receive

Inside the chip: tiles + multiplexer

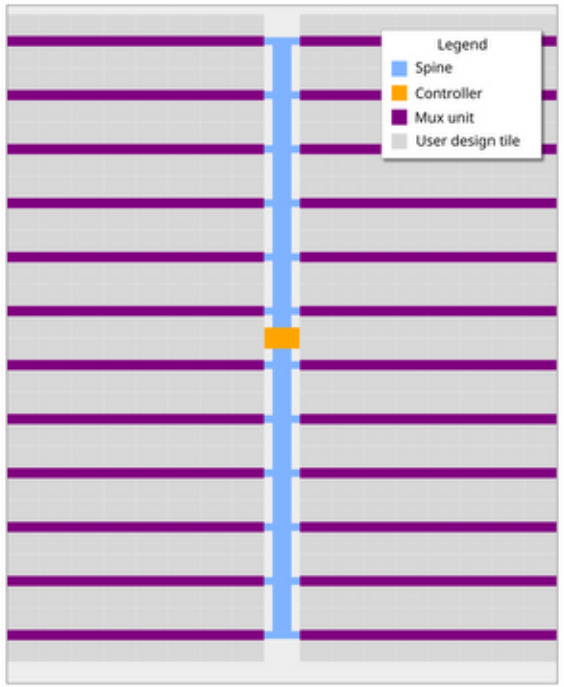


Image: Tiny Tapeout, tt-multiplexer documentation

What you receive: chip on breakout + demo board

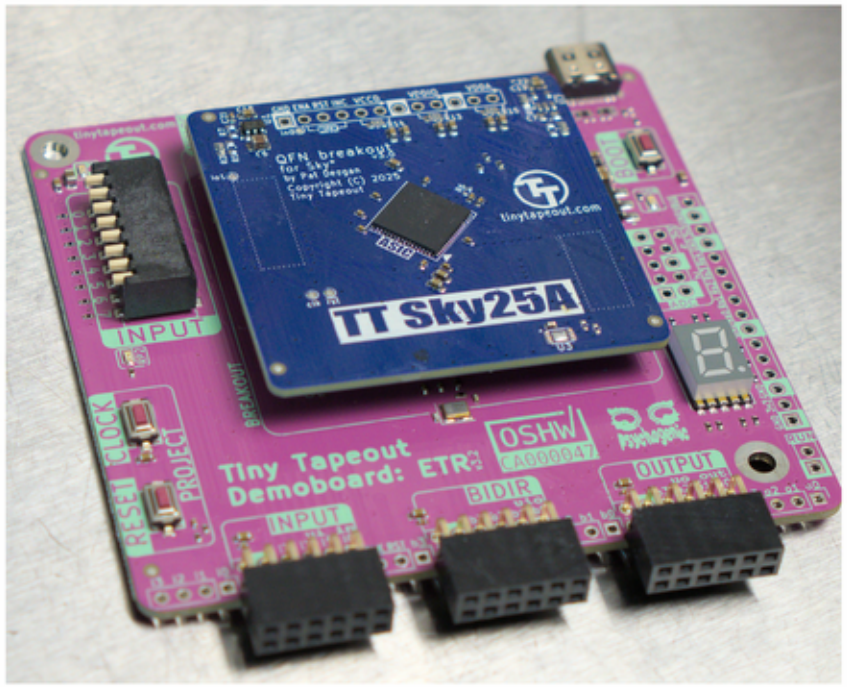


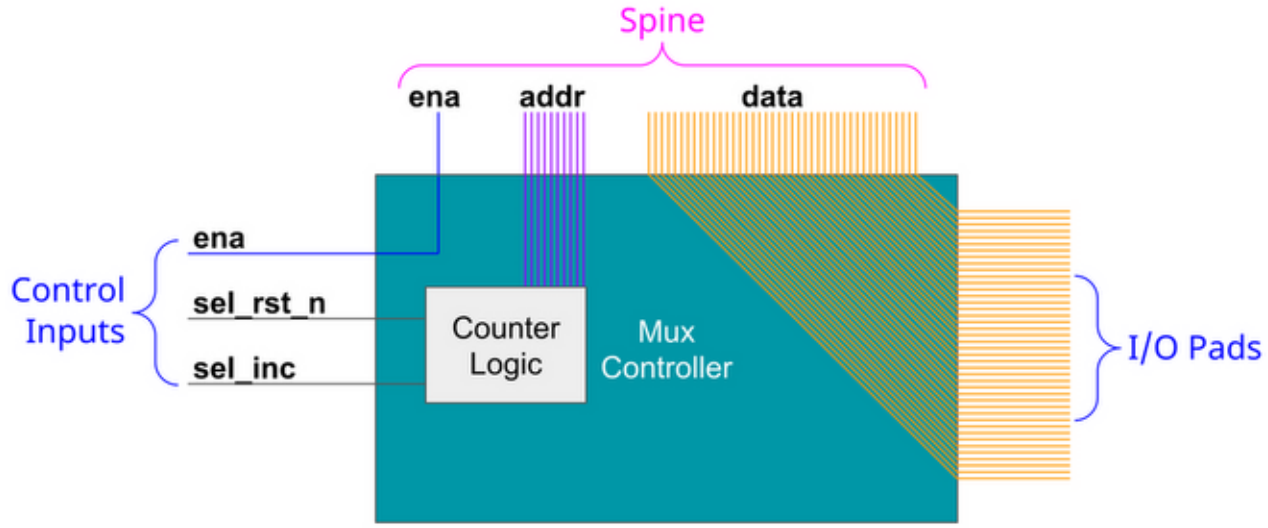
Photo: Tiny Tapeout, ETR demo board with TTSKY25a breakout

KEY POINTS

- A controller selects one design at a time; only that design connects to the shared IO pins
- Inactive designs get clock, reset and inputs tied to 0, and their outputs disconnected
- Every buyer receives a copy of the same chip: ASIC on a breakout board + demo board
- Demo board: RP2350 + MicroPython SDK to select a design, set the clock and inputs, read outputs
- DIP switches, 7-segment display, Pmod headers and USB-C for testing
- For this adder: set ui_in[3:0] (A, B), read uo_out[2:0] (sum, carry)

Selecting your design: the multiplexer

Mux controller: 3 control inputs set the address on the spine



Selecting address 12



reset → 12 increments → enable

Control pins on the breakout board



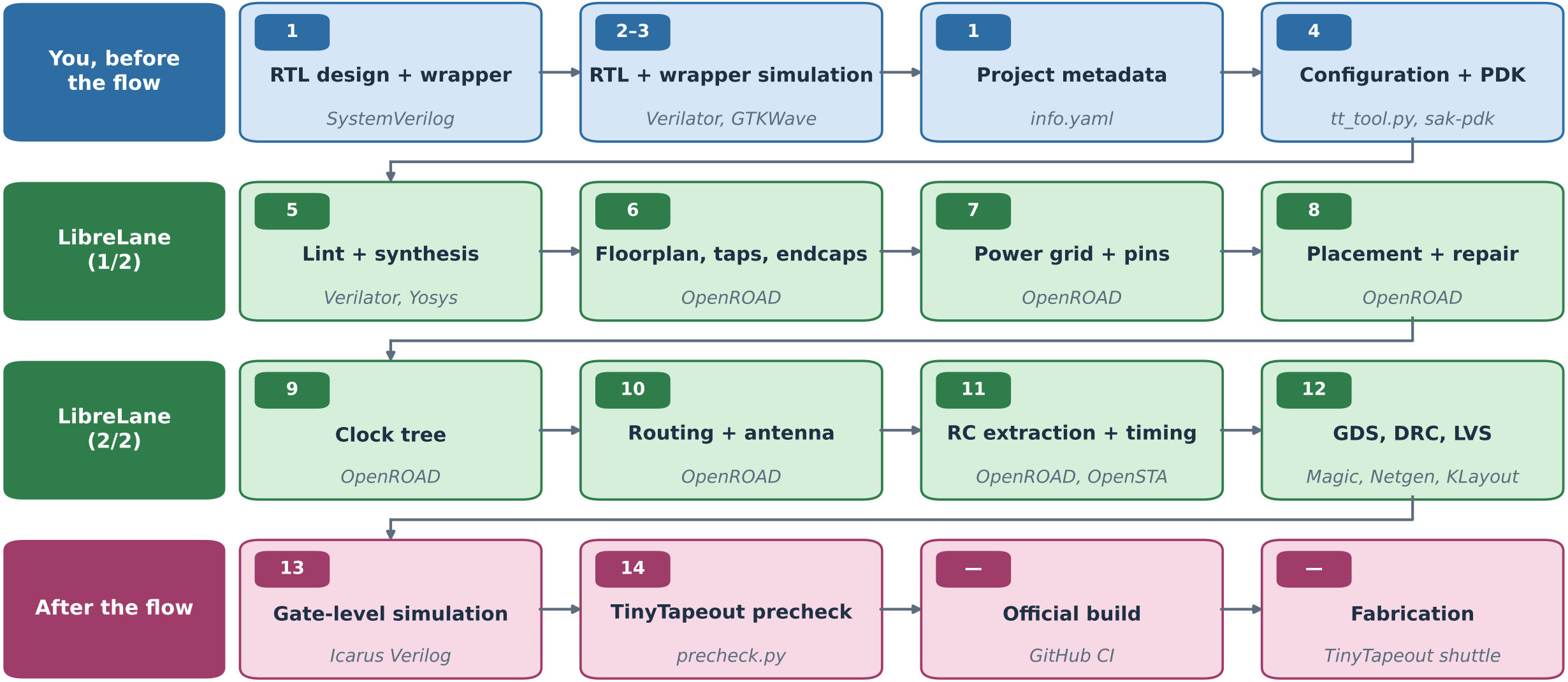
GND · ENA · RST · INC ("MUX")

Diagrams: Tiny Tapeout tt-multiplexer documentation · Photo crop: Tiny Tapeout ETR demo board

KEY POINTS

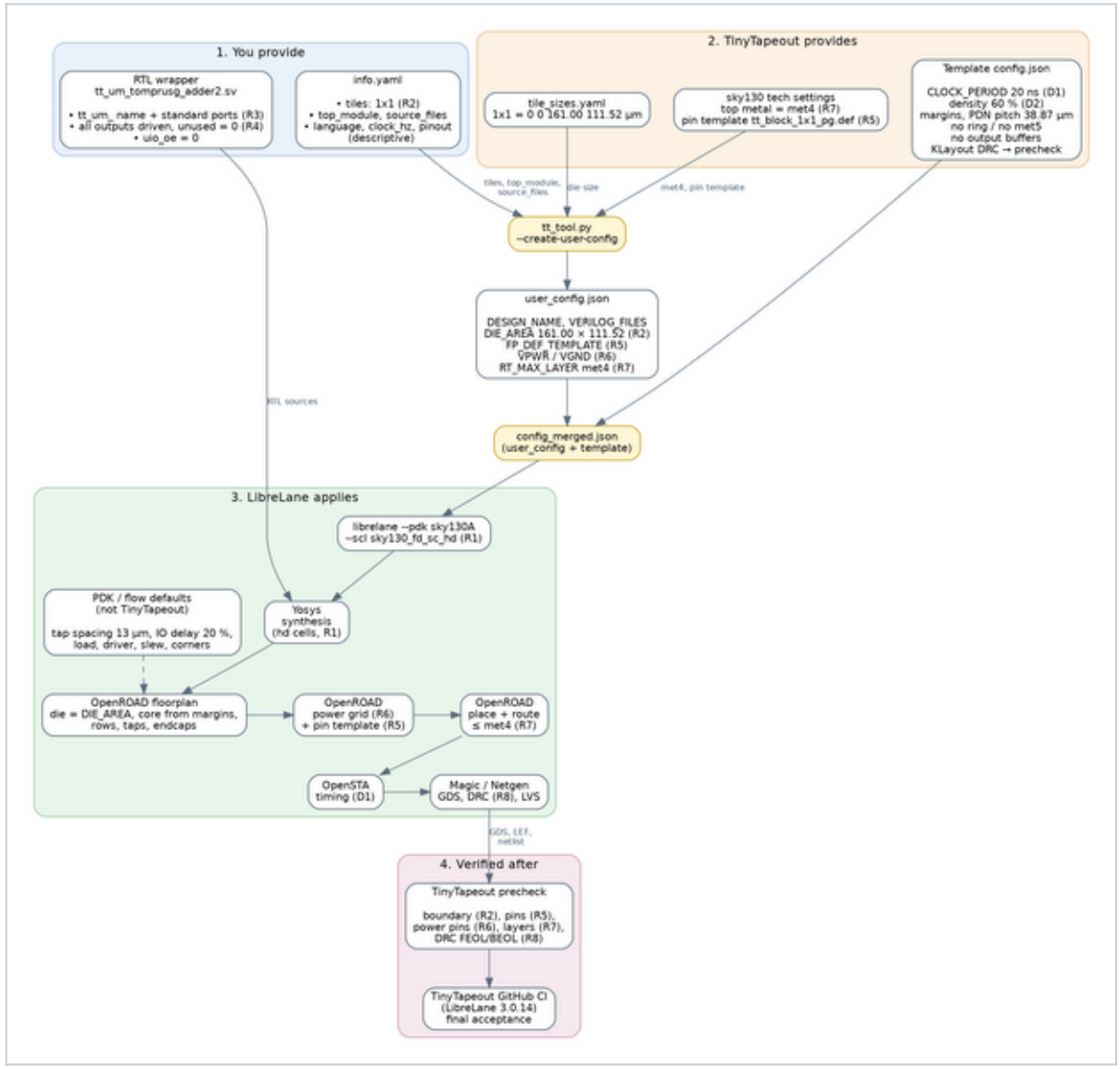
- Built by Tiny Tapeout around the tiles: controller (die centre), spine (vertical bus), mux units (between tile rows)
- Every design has a fixed address 0–511 from its position on the die, listed on the shuttle page
- 3 chip pins reach the controller: ena, sel_rst_n, sel_inc ('MUX' header on the breakout)
- Select: pulse sel_rst_n low (address 0) → pulse sel_inc N times → raise ena; e.g. 12 pulses = design 12
- Only the selected design drives the shared IO pads; all others are isolated
- Normally automatic: the demo board's RP2350 pulses the pins when you pick a project by name
- Your tile's 43 top-edge pins face a mux strip: why the pin template is fixed (R5)

The flow, step by step



Numbers = section numbers in this presentation · LibreLane runs steps 5–12 automatically with one command (≈ 80 internal stages)

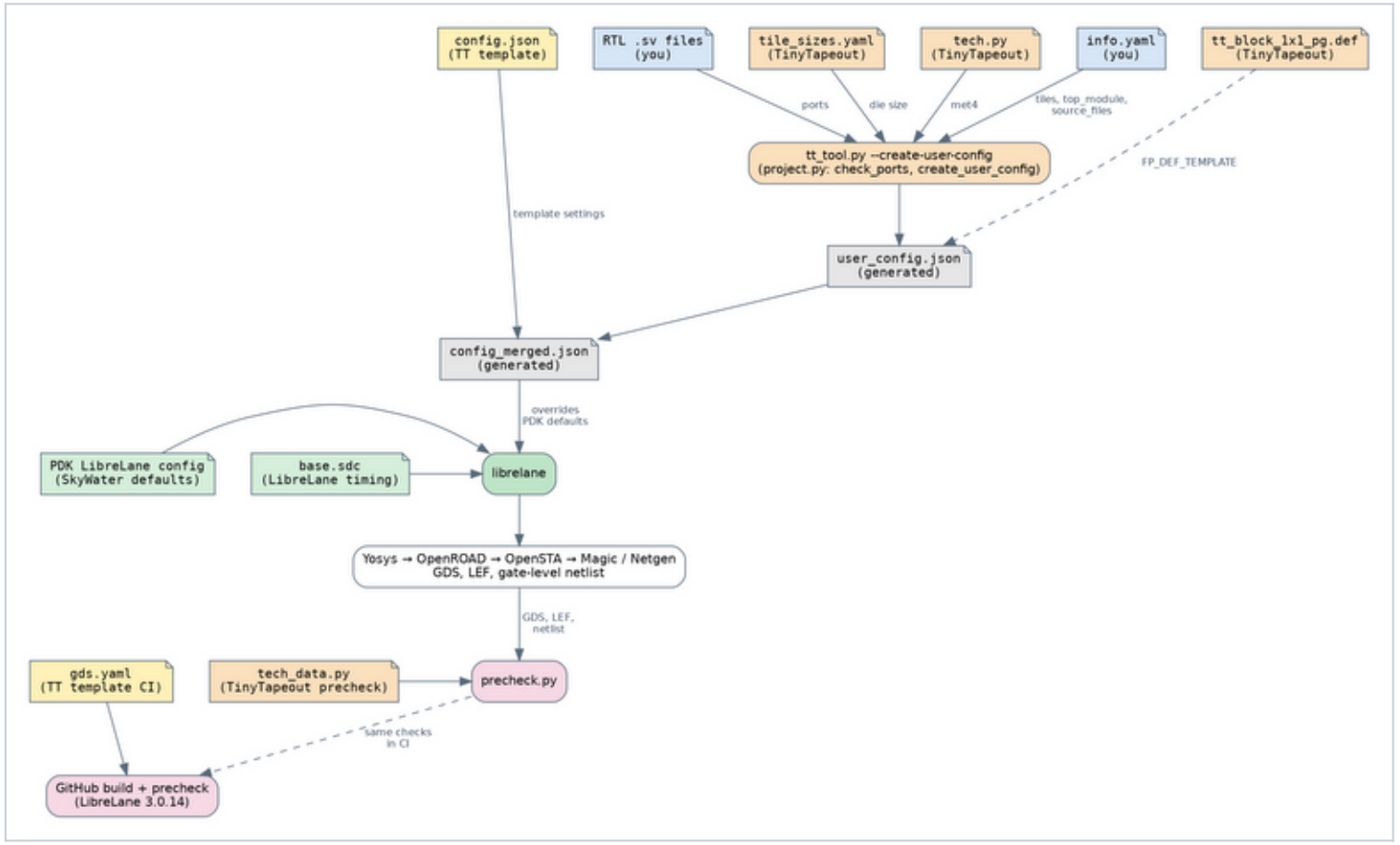
How TinyTapeout constraints enter the flow



KEY POINTS

- You provide: RTL wrapper (tt_um_ name, standard ports, unused outputs = 0) and info.yaml (tiles 1x1, top module, sources)
- TinyTapeout provides: tile table (1x1 = 161.00 × 111.52 µm), sky130 tech settings (met4 max), pin template, template config.json
- tt_tool.py --create-user-config → user_config.json → merged with config.json → config_merged.json
- LibreLane passes the settings to Yosys, OpenROAD, OpenSTA, Magic, Netgen
- Precheck and the GitHub build verify the result
- **Nothing is drawn by hand: every constraint enters as a setting**

The constraint files and who owns them



KEY POINTS

- Yours: info.yaml, RTL .sv files
- TinyTapeout support tools: tile_sizes.yaml, tech.py, project.py (port check), tt_block_1x1_pg.def (pin template)
- TinyTapeout template: config.json (flow recipe), gds.yaml (official CI)
- Generated: user_config.json, config_merged.json (the file LibreLane runs)
- Not TinyTapeout: SkyWater PDK defaults (taps, load, corners), LibreLane base.sdc (timing)
- **Priority: PDK defaults < template < user_config, e.g. PDK met5 → TinyTapeout met4**

Adder core (SystemVerilog)

adder2.sv: unsigned 2-bit adder core (SystemVerilog)

```
1  `default_nettype none
2
3  module adder2 (
4      input  logic [1:0] a,
5      input  logic [1:0] b,
6      output logic [1:0] sum,
7      output logic       carry
8  );
9
10     assign {carry, sum} = {1'b0, a} + {1'b0, b};
11
12 endmodule
13
14 `default_nettype wire
```

KEY POINTS

- Unsigned 2-bit operands a, b → 2-bit sum + carry (results 0–6)
- {carry, sum} = {1'b0, a} + {1'b0, b}: operands zero-extended so the carry falls out of the add
- `default_nettype none: undeclared nets become errors
- Purely combinational: no clock, no flip-flops

TinyTapeout wrapper

tt_um_tomprusg_adder2.sv: TinyTapeout wrapper

```
1  `default_nettype none
2
3  module tt_um_tomprusg_adder2 (
4      input  logic [7:0] ui_in,
5      output logic [7:0] uo_out,
6      input  logic [7:0] uio_in,
7      output logic [7:0] uio_out,
8      output logic [7:0] uio_oe,
9      input  logic       ena,
10     input  logic       clk,
11     input  logic       rst_n
12 );
13
14     logic [1:0] sum;
15     logic       carry;
16
17     // Connect the existing adder to TinyTapeout's input bus.
18     adder2 u_adder (
19         .a     (ui_in[1:0]),
20         .b     (ui_in[3:2]),
21         .sum   (sum),
22         .carry (carry)
23     );
24
25     // Assign every dedicated output bit.
26     assign uo_out = {5'b0, carry, sum};
27
28     // Bidirectional pins are unused and configured as inputs.
29     assign uio_out = 8'b0;
30     assign uio_oe  = 8'b0;
31
32     // Document unused inputs for lint.
33     wire _unused = &{ui_in[7:4], uio_in, ena, clk, rst_n, 1'b0};
34
35 endmodule
36
37 `default_nettype wire
```

KEY POINTS

- Top module tt_um_tomprusg_adder2 with the full standard port list (R3)
- a → ui_in[1:0], b → ui_in[3:2]; sum → uo_out[1:0], carry → uo_out[2]
- uo_out[7:3], uio_out, uio_oe tied to 0: every output driven (R4); uio_oe = 0 means input
- Unused inputs (ui_in[7:4], uio_in, clk, rst_n, ena) collected in _unused for lint
- Both files simulated and fabricated from the same source

info.yaml: project metadata

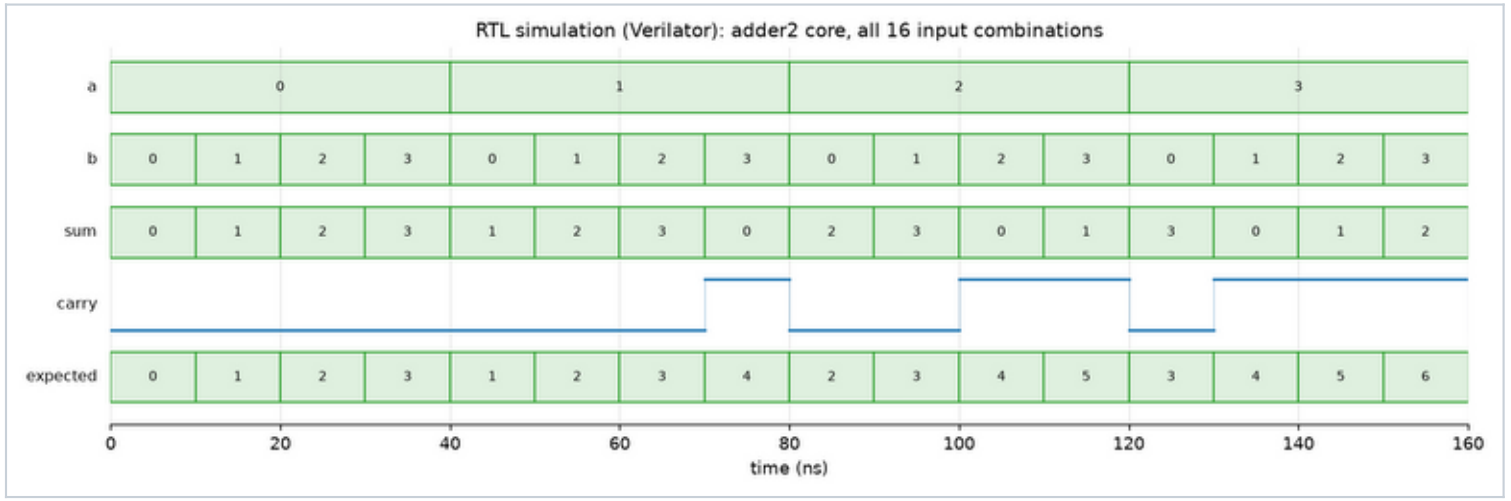
info.yaml: TinyTapeout project metadata (excerpt)

```yaml
project:
  title: "Two-bit unsigned adder"
  description: "Adds two unsigned 2-bit operands and outputs sum and carry."
  language: "SystemVerilog"
  clock_hz: 0
  tiles: "1x1"
  top_module: "tt_um_tomprusg_adder2"
  source_files:
    - "tt_um_tomprusg_adder2.sv"
    - "adder2.sv"
pinout:
  ui[0]: "A bit 0"
  ui[1]: "A bit 1"
  ui[2]: "B bit 0"
  ui[3]: "B bit 1"
  uo[0]: "Sum bit 0"
  uo[1]: "Sum bit 1"
  uo[2]: "Carry"
yaml_version: 6
```

KEY POINTS

- tiles: 1x1 → selects the 161.00 × 111.52 µm die (R2)
- top_module + source_files → tell the tools what to build (R3)
- language SystemVerilog, clock_hz 0 (combinational): descriptive
- pinout: documentation for the datasheet
- Read by tt_tool.py, the precheck and the GitHub build

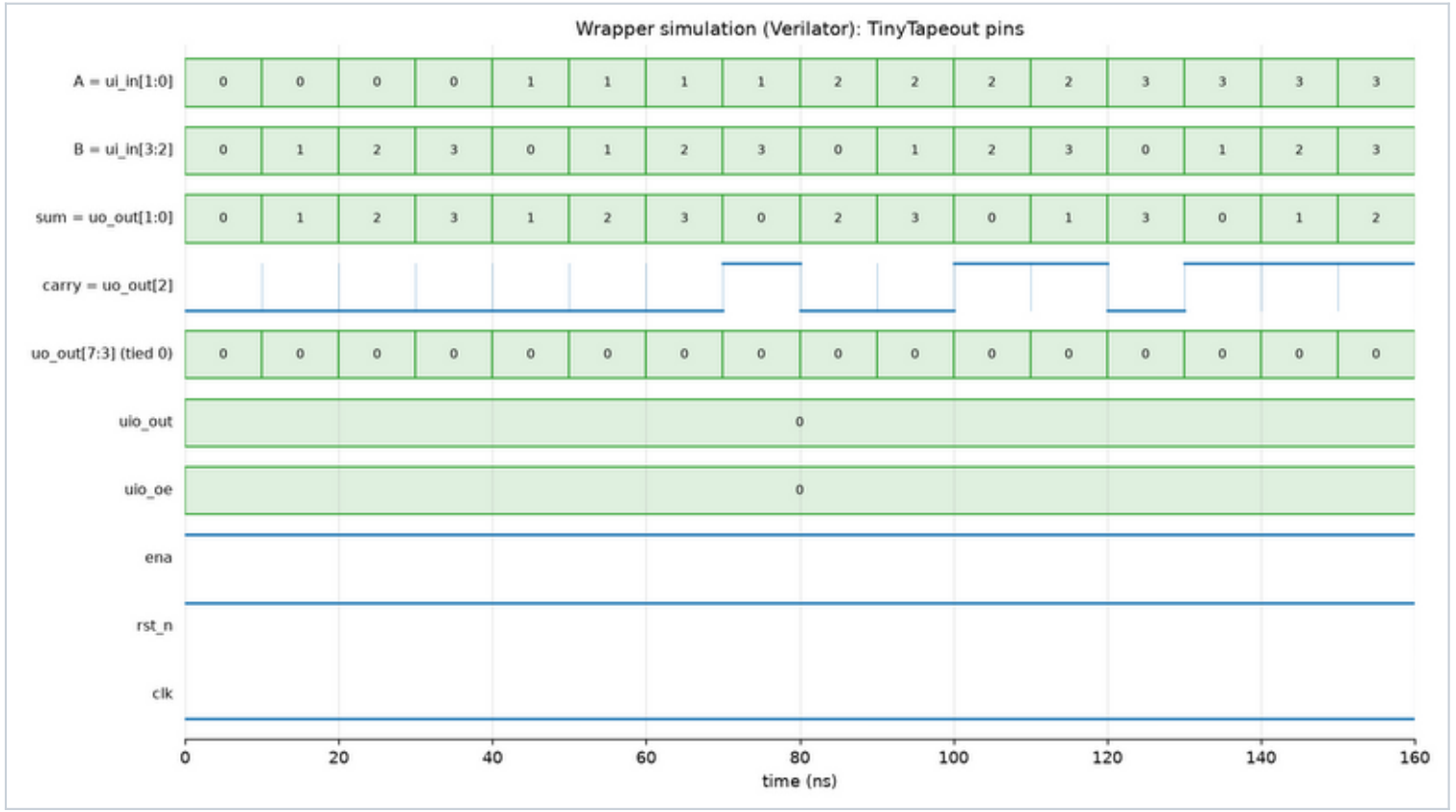
Verilator: exhaustive self-checking test



KEY POINTS

- All 16 combinations of a and b, each checked against a + b
- Stops with $fatal on any mismatch
- -Wall: 0 warnings
- **Result: 16/16 correct, PASS**
- 10 ns between cases is testbench spacing, not a delay or constraint

Verilator: test through the TinyTapeout pins



KEY POINTS

- Same 16 cases through ui_in / uo_out
- Checks the whole 8-bit uo_out bus: upper bits must be 0
- Checks uio_out = 0 and uio_oe = 0 (bidirectionals as inputs)
- Conditions: ena = 1, rst_n = 1, clk = 0
- **Result: 16/16 correct, PASS**

tt_tool.py → user_config.json

user_config.json: generated by tt_tool.py from info.yaml

```
{
  "DESIGN_NAME": "tt_um_tomprusg_adder2",
  "VERILOG_FILES": [
    "dir::tt_um_tomprusg_adder2.sv",
    "dir::adder2.sv"
  ],
  "DIE_AREA": "0 0 161.00 111.52",
  "FP_DEF_TEMPLATE": "dir::../tt/tech/sky130A/def/tt_block_1x1_pg.def",
  "VDD_PIN": "VPWR",
  "GND_PIN": "VGND",
  "RT_MAX_LAYER": "met4"
}
```

KEY POINTS

- Generated, not typed, from info.yaml + TinyTapeout tables
- DIE_AREA 161.00 × 111.52 µm (R2)
- FP_DEF_TEMPLATE: the 1x1 pin template (R5)
- VPWR / VGND power names (R6); RT_MAX_LAYER met4 (R7)
- Merged with template config.json: 20 ns clock reference, 60 % density, fixed margins and power grid
- Ports checked first: tool stops if a required port is missing

config_merged.json: TinyTapeout requirements

config_merged.json: TinyTapeout requirements (excerpt)

```
{
  "DESIGN_NAME": "tt_um_tomprusg_adder2",
  "VERILOG_FILES": [
    "dir::tt_um_tomprusg_adder2.sv",
    "dir::adder2.sv"
  ],
  "DIE_AREA": "0 0 161.00 111.52",
  "FP_DEF_TEMPLATE": "dir::../tt/tech/sky130A/def/tt_block_1x1_pg.def",
  "VDD_PIN": "VPWR",
  "GND_PIN": "VGND",
  "RT_MAX_LAYER": "met4",
  "CLOCK_PORT": "clk",
  "CLOCK_PERIOD": 20,
  "PL_TARGET_DENSITY_PCT": 60
}
```

DESIGN_NAME
Top module tt_um_tomprusg_adder2: the unique tt_um_ name (R3).

VERILOG_FILES
The two RTL sources given to the linter and Yosys.

DIE_AREA 0 0 161.00 111.52
The 1x1 tile in µm, from TinyTapeout's tile table (R2).

FP_DEF_TEMPLATE
TinyTapeout pin template: pins copied to exact positions, size and layer (R5).

VDD_PIN / GND_PIN
Power net names VPWR / VGND, used by the power grid and LVS (R6).

RT_MAX_LAYER met4
Highest layer the router may use; met5 is reserved for the chip (R7).

CLOCK_PORT / CLOCK_PERIOD
20 ns reference clock on clk; for this adder only a timing reference (D1).

PL_TARGET_DENSITY_PCT 60
Max local cell density during placement: how tightly cells are packed (D2).

This file
30 settings: generated user_config.json merged over the template config.json.

config_merged.json: template settings explained

config_merged.json: template settings (excerpt)

```
{
  "FP_SIZING": "absolute",
  "TOP_MARGIN_MULT": 1,
  "BOTTOM_MARGIN_MULT": 1,
  "LEFT_MARGIN_MULT": 6,
  "RIGHT_MARGIN_MULT": 6,
  "FP_PDN_VPITCH": 38.87,
  "FP_PDN_MULTILAYER": 0,
  "FP_IO_HLENGTH": 2,
  "FP_IO_VLENGTH": 2,
  "PL_RESIZER_HOLD_SLACK_MARGIN": 0.1,
  "GRT_RESIZER_HOLD_SLACK_MARGIN": 0.05,
  "GRT_ALLOW_CONGESTION": 1,
  "DESIGN_REPAIR_BUFFER_OUTPUT_PORTS": 0,
  "RUN_CTS": 1,
  "RUN_LINTER": 1,
  "LINTER_INCLUDE_PDK_MODELS": 1,
  "RUN_KLAYOUT_DRC": 0,
  "RUN_KLAYOUT_XOR": 0,
  "MAGIC_DEF_LABELS": 0,
  "MAGIC_WRITE_LEF_PINONLY": 1
}
```

FP_SIZING absolute
Die size is taken literally from DIE_AREA instead of being computed from cell area.

***_MARGIN_MULT 1 / 1 / 6 / 6**
Core inset: 1 row height top/bottom (2.72 µm), 6 sites left/right (2.76 µm).

FP_PDN_VPITCH 38.87
Distance (µm) between met4 power-stripe pairs; gives 4 pairs across the core.

FP_PDN_MULTILAYER 0
Power grid on met4 only: no met5 straps, no ring. The chip grid connects from above.

FP_IO_HLENGTH / VLENGTH 2
Pin length (µm) for LibreLane's own IO placer; here the TinyTapeout template pins are used instead.

PL_ / GRT_RESIZER_HOLD_SLACK_MARGIN
Extra hold margin (0.1 / 0.05 ns) the resizer aims for after placement / global routing.

GRT_ALLOW_CONGESTION 1
Global routing may continue with overfull regions; detailed routing must still be clean.

DESIGN_REPAIR_BUFFER_OUTPUT_PORTS 0
No automatic buffer on every output; buffers only where a violation needs them.

RUN_CTS 1
Clock-tree step enabled; nothing to build here (no flip-flops).

RUN_LINTER / LINTER_INCLUDE_PDK_MODELS
Verilator lint before synthesis, with cell models so instantiated cells are recognized.

RUN_KLAYOUT_DRC / XOR 0
Skipped in the flow to save time; KLayout DRC is run by the TinyTapeout precheck.

MAGIC_DEF_LABELS 0
Pin labels from the DEF are not copied into Magic's layout; avoids LVS label problems.

MAGIC_WRITE_LEF_PINONLY 1
The exported LEF shows only pin shapes, not the metal of nets connected to them.

Final SDC + resolved.json: what the tools actually used

Final SDC + resolved.json (written by LibreLane): key lines

```
# ---- final SDC: timing constraints the tools applied ----
create_clock -name clk -period 20.0000 [get_ports {clk}]
set_clock_transition 0.1500 [get_clocks {clk}]
set_clock_uncertainty 0.2500 clk
set_timing_derate -early 0.9500
set_timing_derate -late 1.0500
set_max_transition 0.7500 [current_design]
set_max_capacitance 0.2000 [current_design]
set_max_fanout 10.0000 [current_design]
set_driving_cell -lib_cell sky130_fd_sc_hd__inv_2 -pin {Y} ... [get_ports {ui_in[0]}]
   (same for every input)
set_input_delay 4.0000 -clock [get_clocks {clk}] -add_delay [get_ports {ui_in[0]}]
set_output_delay 4.0000 -clock [get_clocks {clk}] -add_delay [get_ports {uo_out[0]}]
set_load -pin_load 0.0334 [get_ports {uo_out[0]}]
   (same for every input / output)

# ---- resolved.json: 421 final settings (excerpt) ----
STD_CELL_LIBRARY = sky130_fd_sc_hd
PLACE_SITE = unithd
RT_MAX_LAYER = met4
WELLTAP_CELL = sky130_fd_sc_hd__tapvpwrvgnd_1
ENDCAP_CELL = sky130_fd_sc_hd__decap_3
FP_TAPCELL_DIST = 13
PDN_VPITCH = 38.87
PDN_VWIDTH = 1.6
PDN_RAIL_WIDTH = 0.48
IO_DELAY_CONSTRAINT = 20
OUTPUT_CAP_LOAD = 33.442
TIME_DERATING_CONSTRAINT = 5
DEFAULT_CORNER = nom_tt_025C_1v80
STA_CORNERS = 9 corners: {nom,min,max} x {tt_025C_1v80, ss_100C_1v60, ff_n40C_1v95}
```

`create_clock -period 20`
Reference clock on clk; every input-to-output path is timed against it.

`set_input_delay / set_output_delay 4.0`
20 % of the period assumed outside the tile on each pin; 11.75 ns remain inside.

`set_driving_cell inv_2 · set_load 0.0334`
Inputs driven by an inv_2; each output drives 33.4 fF (an inv_16 input).

`set_clock_uncertainty 0.25 · transition 0.15`
Margin for clock jitter/skew; assumed clock-edge time in ns.

`set_timing_derate 0.95 / 1.05`
All delays varied ±5 % to cover on-chip variation.

`set_max_transition / capacitance / fanout`
Electrical limits on every net: 0.75 ns edges, 0.2 pF load, 10 loads.

`resolved.json`
All 421 final settings: yours + template + SkyWater PDK defaults + derived values.

`PLACE_SITE, WELLTAP/ENDCAP_CELL, FP_TAPCELL_DIST`
PDK defaults: 0.46 × 2.72 µm site, tap and endcap cells, taps ≤ 13 µm apart.

`RT_MAX_LAYER = met4`
PDK default is met5: proof that TinyTapeout's setting overrides the PDK.

`STA_CORNERS`
9 timing corners: tt / ss / ff × min / nom / max wire parasitics.

The PDK: LibreLane's third input

PDK selection: sak-pdk (IIC) + LibreLane PDK options

```
$ sak-pdk sky130A                       # IIC helper: select the PDK in this shell
PDK_ROOT=/foss/pdks
PDK=sky130A
PDKPATH=/foss/pdks/sky130A
STD_CELL_LIBRARY=sky130_fd_sc_hd
SPICE_USERINIT_DIR=/foss/pdks/sky130A/libs.tech/ngspice
KLAYOUT_PATH=/headless/.klayout:/foss/pdks/sky130A/libs.tech/klayout

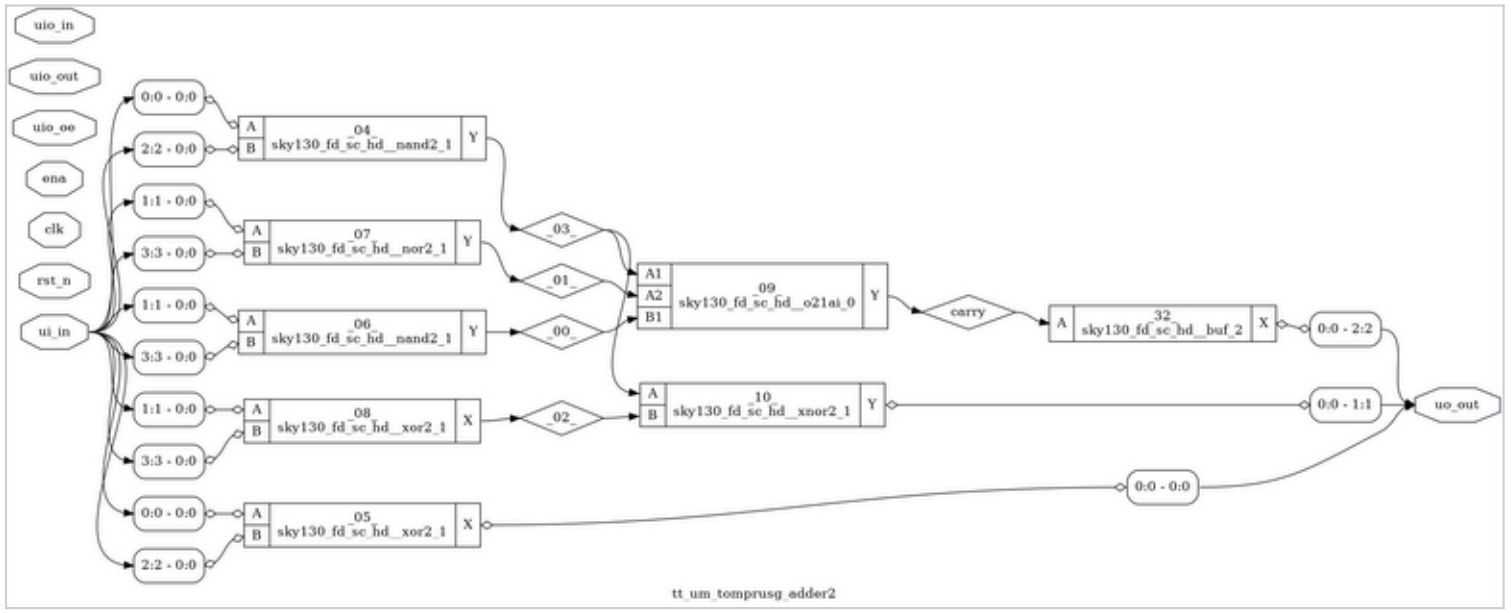
$ librelane --manual-pdk --pdk-root $PDK_ROOT --pdk sky130A \
            --scl sky130_fd_sc_hd --run-tag full config_merged.json

# sky130A PDK  (open_pdks 1689ac3f)
libs.ref/sky130_fd_sc_hd/       standard-cell library views
   lib/        Liberty: function, timing, power   -> Yosys, OpenSTA
   lef/ techlef/  abstracts, layers, sites         -> OpenROAD
   gds/  mag/  maglef/  full cell layouts          -> Magic, KLayout
   spice/  cdl/   transistor netlists              -> Netgen LVS
   verilog/    cell models                         -> gate-level sim
libs.tech/                      tool setup files
   librelane/  flow defaults (taps, load, corners) -> LibreLane
   magic/  netgen/  klayout/   DRC, extraction, LVS rules
```

KEY POINTS

- LibreLane gets three inputs: config_merged.json, the RTL, and the PDK, which holds all the technology data
- sak-pdk sky130A: IIC helper that selects the PDK in the current shell (PDK_ROOT, PDK, STD_CELL_LIBRARY …)
- Needed in every new shell: IIC starts with IHP sg13g2 selected; our first synthesis failed for this reason
- LibreLane options: --pdk sky130A --scl sky130_fd_sc_hd (R1); --manual-pdk --pdk-root uses the installed copy
- Tools read the library views: Liberty (Yosys, OpenSTA), LEF (OpenROAD), GDS (Magic), SPICE (LVS), Verilog (gate-level sim)
- Local PDK open_pdks 1689ac3f; Tiny Tapeout CI uses 8afc8346 with LibreLane 3.0.14

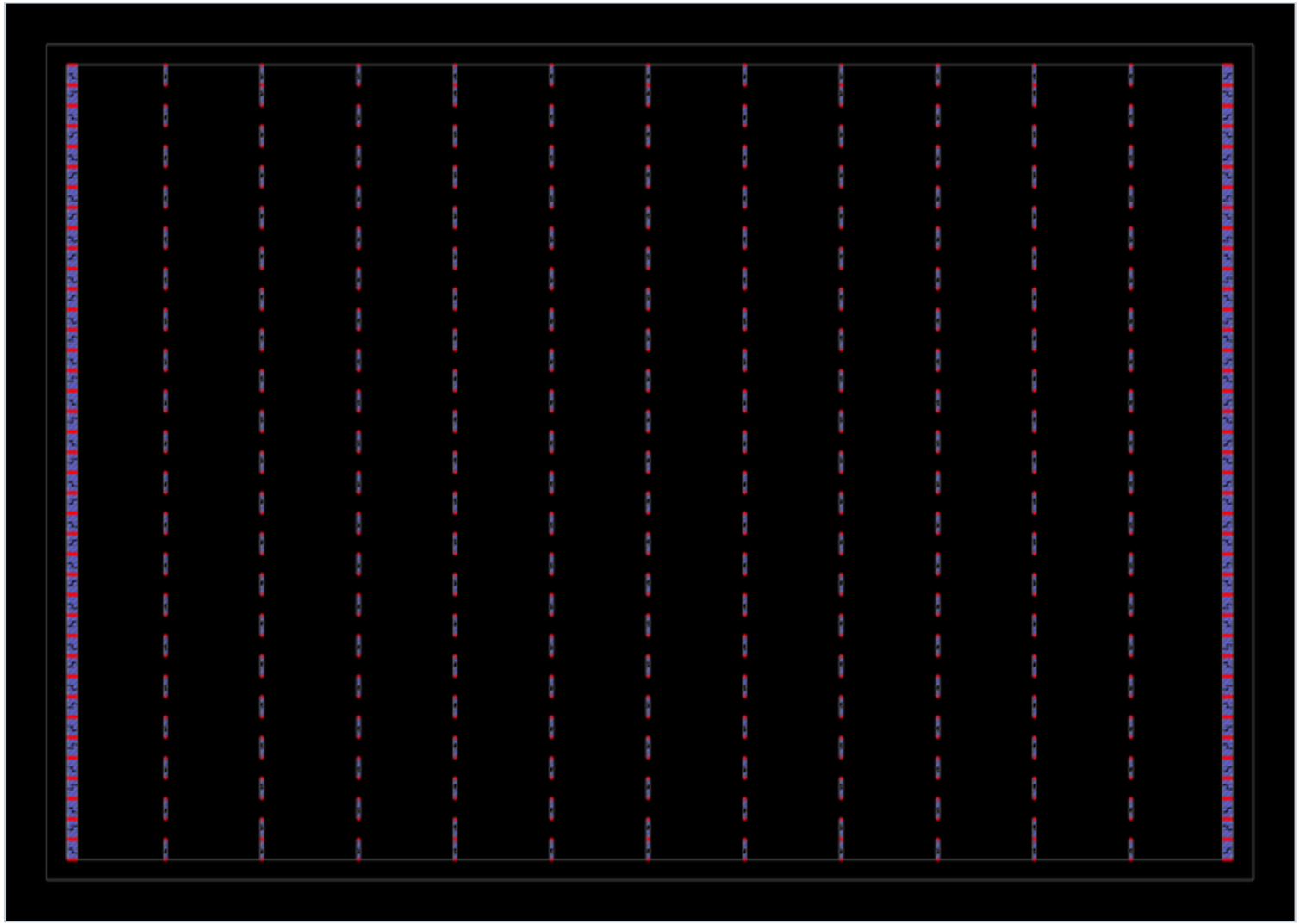
Yosys: RTL → SkyWater standard cells



KEY POINTS

- Maps RTL to sky130_fd_sc_hd cells using only their Liberty models (function, area, timing)
- 29 cells, 126.4 µm²: 7 logic + 1 buffer + 21 tie cells
- Inverting gates preferred (NAND, NOR, OAI = one CMOS stage)
- carry = o21ai: a1·b1 + c0·(a1+b1); sum1 = xnor using inverted c0
- 0 flip-flops, 0 latches, 0 lint warnings
- **Command: librelane … config_merged.json (runs the whole flow)**

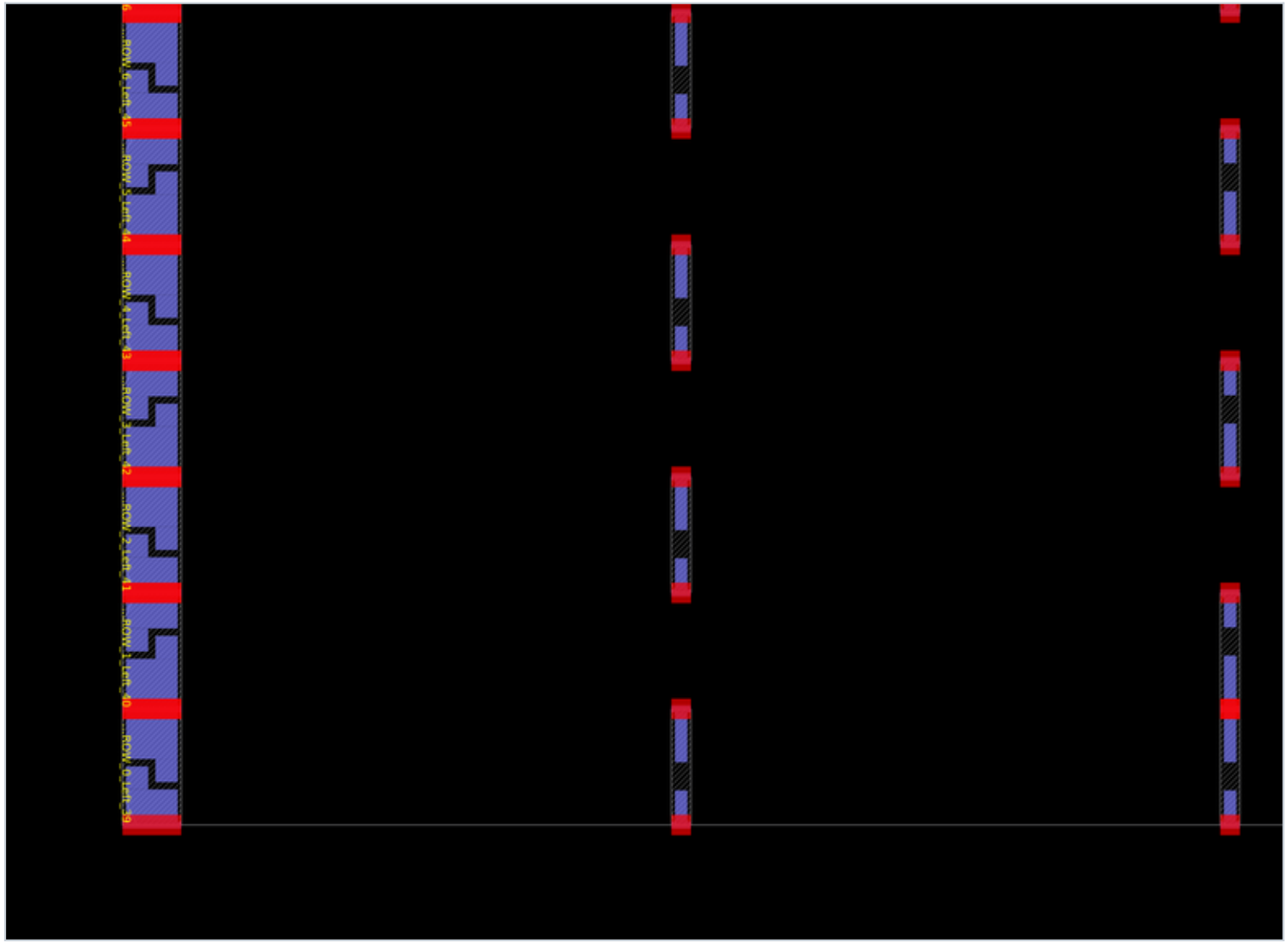
OpenROAD: die, core and rows



KEY POINTS

- Die 161.00 × 111.52 µm from DIE_AREA (TinyTapeout)
- Core inset: 6 sites left/right (2.76 µm), 1 row top/bottom (2.72 µm) (template)
- Site unithd = 0.46 × 2.72 µm (PDK) → 39 rows × 338 sites (calculated)
- Tile = exactly 41 rows × 350 sites: TinyTapeout chose it to fit the grid
- **Utilization 0.77 %: fits with huge margin**

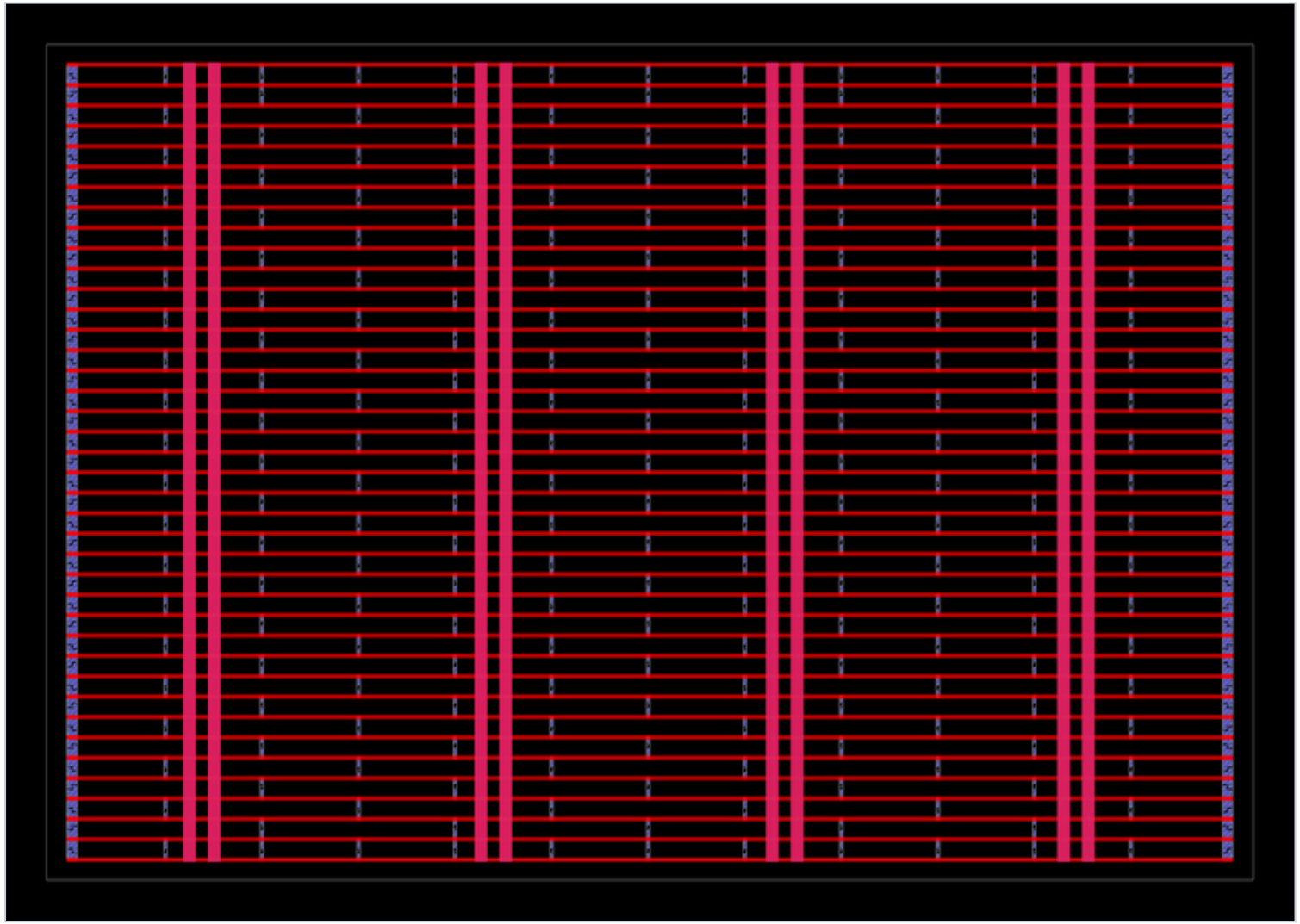
Well taps and endcaps



KEY POINTS

- sky130 hd cells are 'tapless': no well contacts inside logic cells
- Tap cell = n-well tie to VPWR + substrate tie to VGND (no transistors)
- Latch-up rule: taps ≤ 13 µm apart → checkerboard, 225 taps
- Placed everywhere, before placement: logic location unknown, filler wells also need ties
- Endcaps (decap_3, 78) close wells/implants at row ends; not passivation

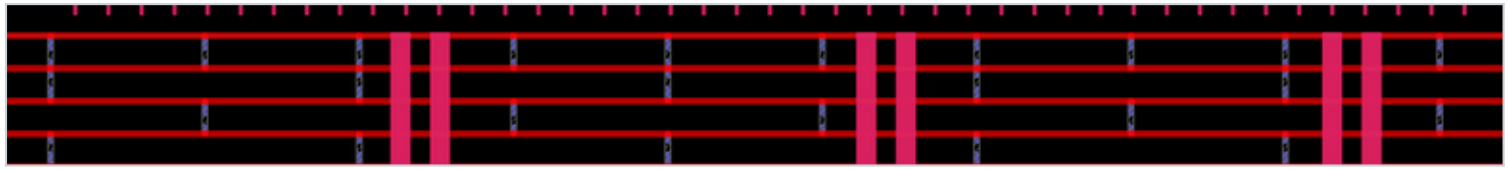
OpenROAD: VPWR / VGND network



KEY POINTS

- met1 rails (0.48 µm) on every row boundary: cells connect by abutment
- 4 pairs of met4 stripes, 1.6 µm wide, every 38.87 µm (template pitch)
- Count = whatever fits the core, not what the logic needs (IR drop only 3.3 µV)
- Via stacks met1 → met4 where stripes cross rails
- No ring, no met5: the chip-level grid connects from above (R7)
- Built before placement: fixed obstacles, cells need the rails

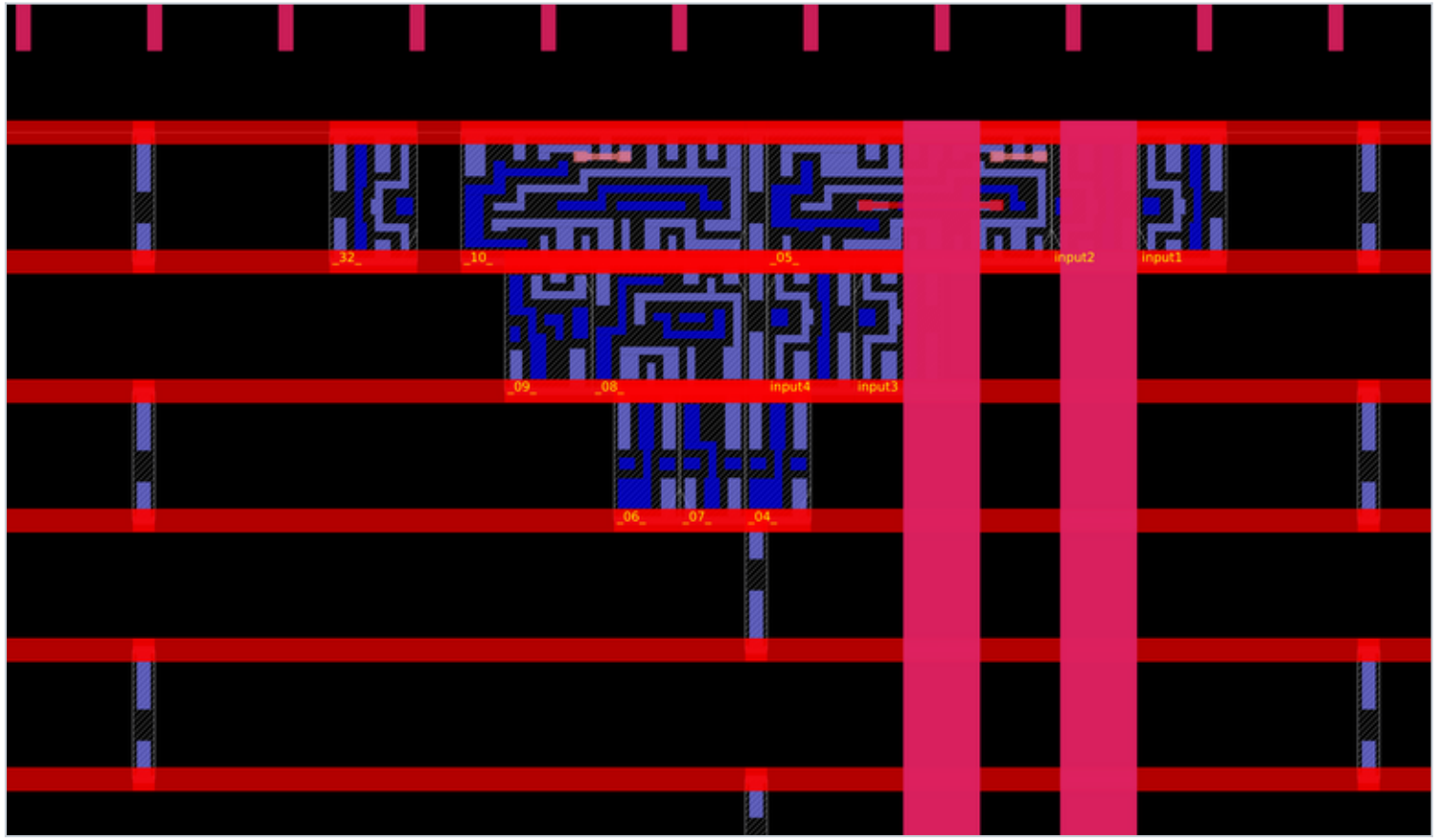
Fixed TinyTapeout pin template



KEY POINTS

- 43 signal pins copied from tt_block_1x1_pg.def, not chosen by the tool (R5)
- All on the top edge, met4, 0.3 × 1.0 µm, 2.76 µm pitch
- Order left → right: uio_oe, uio_out, uo_out, uio_in, ui_in, rst_n, clk, ena
- Inputs used: ui_in[0..3] at x ≈ 130–138 µm; outputs uo_out[0..2] at x ≈ 89–94 µm
- Power pins: one met4 stripe pair, created by the power-grid step (R6)

OpenROAD: global + detailed placement



KEY POINTS

- Optimization, not trial-and-error: minimize total wire length (HPWL) under 60 % density
- Nets = springs, cells = repelling charges; pins and taps are fixed anchors
- Not timing-driven here: delay improved only indirectly via short wires
- Cluster near the inputs: on a straight input→output line every position has equal wire length
- HPWL 326.6 → 294.5 µm after detailed placement; all cells legal on sites

Design repair: buffers and drive strength

OpenROAD: global placement, design repair and detailed placement

```
[INFO GPL-0006] Number of instances:              332
[INFO GPL-0007] Movable instances:                 29
[INFO GPL-0008] Fixed instances:                  303
[INFO GPL-0016] Core area:                  16493.318 um^2
[INFO GPL-0007] ---- Execute Nesterov Global Placement.
[INFO RSZ-0027] Inserted 4 sky130_fd_sc_hd__buf_2 input buffers.
[INFO RSZ-0034] Found 2 slew violations.
[INFO RSZ-0039] Resized 2 instances.
[INFO RSZ-0038] Inserted 2 buffers in 2 nets.
[INFO RSZ-0042] Inserted 21 tie sky130_fd_sc_hd__conb_1 instances.
[INFO DPL-0020] Mirrored 29 instances
[INFO DPL-0021] HPWL before             326.6 u
[INFO DPL-0022] HPWL after              294.5 u
[INFO DPL-0023] HPWL delta               -9.8 %
```

KEY POINTS

- 4 input buffers (buf_2 = two inverters → non-inverting), one per used ui_in pin
- Purpose: one small known load for the chip wiring, strong local drive inside
- 2 slew violations on long output wires → _05_/_10_ upsized _1 → _2, wire5/wire6 added
- Drive strength = output-transistor width: buf_2 output 1.3 µm n / 2.0 µm p (2.5 × buf_1)
- 21 tie cells placed under their output pins; cells 29 → 35

TritonCTS: nothing to do

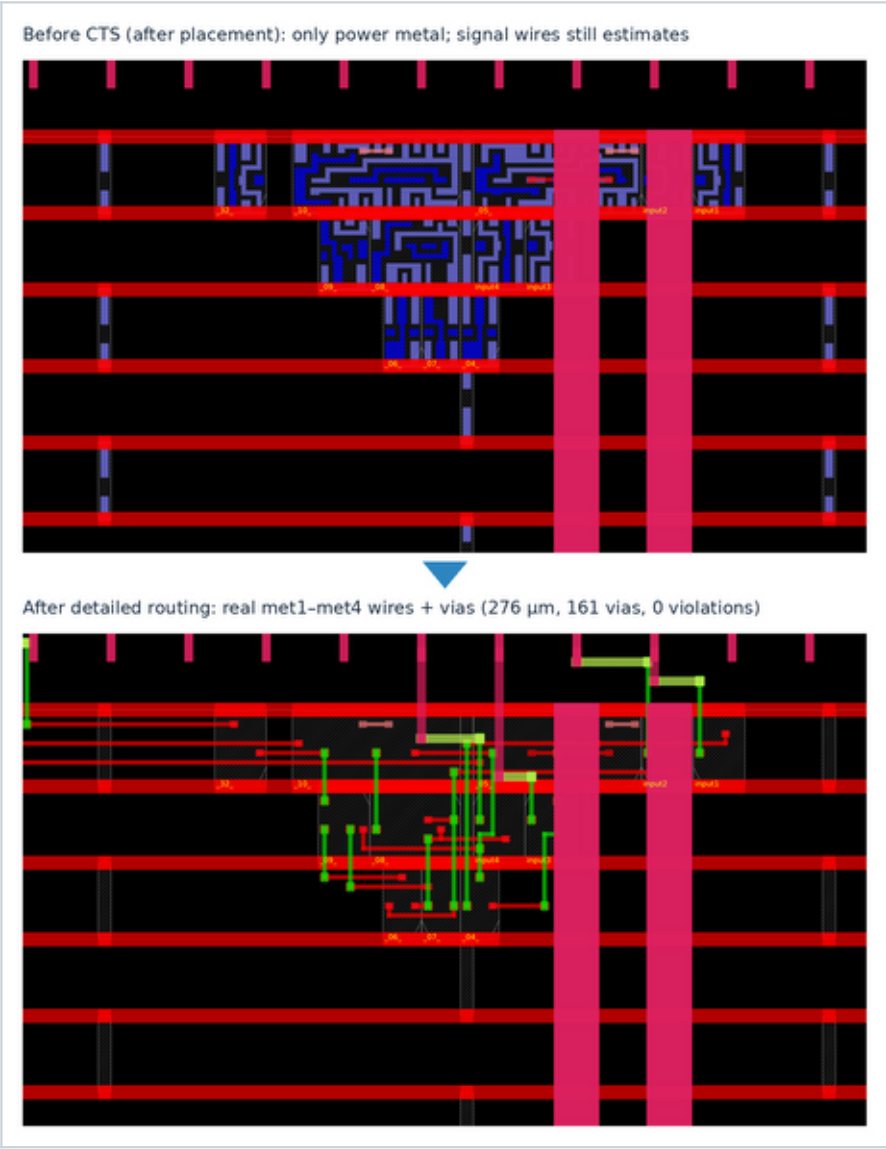
OpenROAD TritonCTS: no clock sinks in a combinational design

```
[INFO CTS-0007] Net "clk" found for clock "clk".
[WARNING CTS-0041] Net "clk" has 0 sinks. Skipping...
[WARNING CTS-0083] No clock nets have been found.
[INFO CTS-0008] TritonCTS found 0 clock nets.
[WARNING CTS-0082] No valid clock nets in the design.
[INFO RSZ-0098] No setup violations found
[INFO RSZ-0033] No hold violations found.
```

KEY POINTS

- CTS builds a balanced buffer tree to every flip-flop clock pin
- clk has 0 sinks: the adder has no flip-flops
- 0 clock buffers inserted
- Post-CTS repair: no setup or hold violations

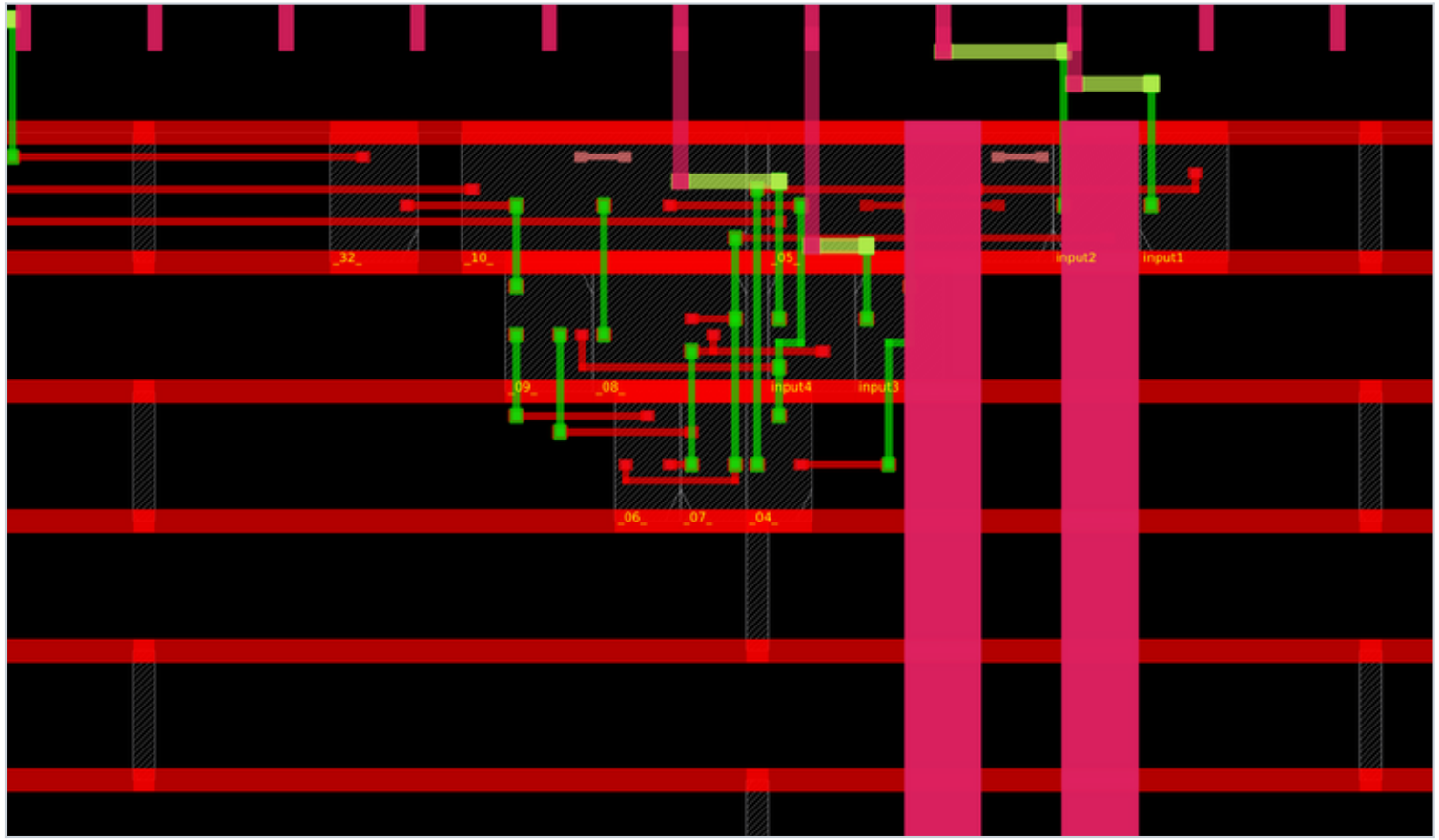
When and how the wires are drawn



KEY POINTS

- No signal routing before CTS: only power metal (met1 rails, met4 stripes) exists after placement
- During placement, timing uses estimated wire length/RC; the placer asks the global router only to predict congestion
- CTS: clock buffers placed and clock nets routed first, with priority and wider wires (empty here: no flip-flops)
- Global routing: die split into a coarse grid (GCells); each net gets regions + layers, not exact wires (max congestion 0.26 %)
- Detailed routing (TritonRoute): real metal + vias on tracks under DRC rules → 276 µm, 161 vias, 0 violations
- **Order: power grid → placement → clock tree → global → detailed routing**

OpenROAD: global + detailed routing



KEY POINTS

- Global routing plans regions/layers; TritonRoute draws real metal + vias
- Preferred directions alternate: met1 H, met2 V, met3 H, met4 V
- Wire length 276 µm: met1 119, met2 52, met3 41, met4 64, met5 0 (R7)
- 161 vias; 0 routing violations; 0 antenna violations
- 15 disconnected pins = exactly the unused inputs (0 critical)

OpenSTA: worst setup path

OpenSTA: worst setup path (max_ss_100C_1v60)

```
Startpoint: ui_in[1] (input port clocked by clk)
Endpoint: uo_out[1] (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout         Cap        Slew       Delay        Time   Description
-----------------------------------------------------------------------------
                                    0.000000    0.000000   clock clk (rise edge)
                                    0.000000    0.000000   clock network delay (propagated)
                                    4.000000    4.000000 v input external delay
     1    0.002517    0.017385    0.008811    4.008811 v ui_in[1] (in)
                                                         ui_in[1] (net)
                     0.017385    0.000000    4.008811 v input2/A (sky130_fd_sc_hd__buf_2)
     3    0.011811    0.079929    0.254679    4.263490 v input2/X (sky130_fd_sc_hd__buf_2)
                                                         net2 (net)
                     0.079934    0.000742    4.264232 v _08_/A (sky130_fd_sc_hd__xor2_1)
     1    0.008832    0.355706    0.397099    4.661331 ^ _08_/X (sky130_fd_sc_hd__xor2_1)
                                                         _02_ (net)
                     0.355706    0.000711    4.662042 ^ _10_/B (sky130_fd_sc_hd__xnor2_2)
     1    0.006712    0.219877    0.320962    4.983005 ^ _10_/Y (sky130_fd_sc_hd__xnor2_2)
                                                         net5 (net)
                     0.219877    0.000420    4.983425 ^ wire5/A (sky130_fd_sc_hd__clkbuf_4)
     1    0.033936    0.177905    0.415858    5.399282 ^ wire5/X (sky130_fd_sc_hd__clkbuf_4)
                                                         uo_out[1] (net)
                     0.177923    0.001967    5.401249 ^ uo_out[1] (out)
                                             5.401249   data arrival time

                                   20.000000   20.000000   clock clk (rise edge)
                                    0.000000   20.000000   clock network delay (propagated)
                                   -0.250000   19.750002   clock uncertainty
                                    0.000000   19.750002   clock reconvergence pessimism
                                   -4.000000   15.750001   output external delay
                                               15.750001   data required time
-----------------------------------------------------------------------------
                                               15.750001   data required time
                                               -5.401249   data arrival time
-----------------------------------------------------------------------------
                                               10.348751   slack (MET)
```

KEY POINTS

- Static: every path, every corner, no test vectors; Liberty delays + extracted wire RC
- Model: external flops launch/capture; 4 ns outside before and after the tile
- Setup (too late?): required 15.75 ns − arrival 5.40 ns = +10.35 ns
- Hold (too early?): fastest path + IO delays vs capture edge = +7.99 ns
- Worst path ui_in[1] → buf → xor → xnor → wire5 → uo_out[1]: 1.40 ns inside the tile
- **0 setup, hold, slew, cap violations**

Signoff in 9 corners

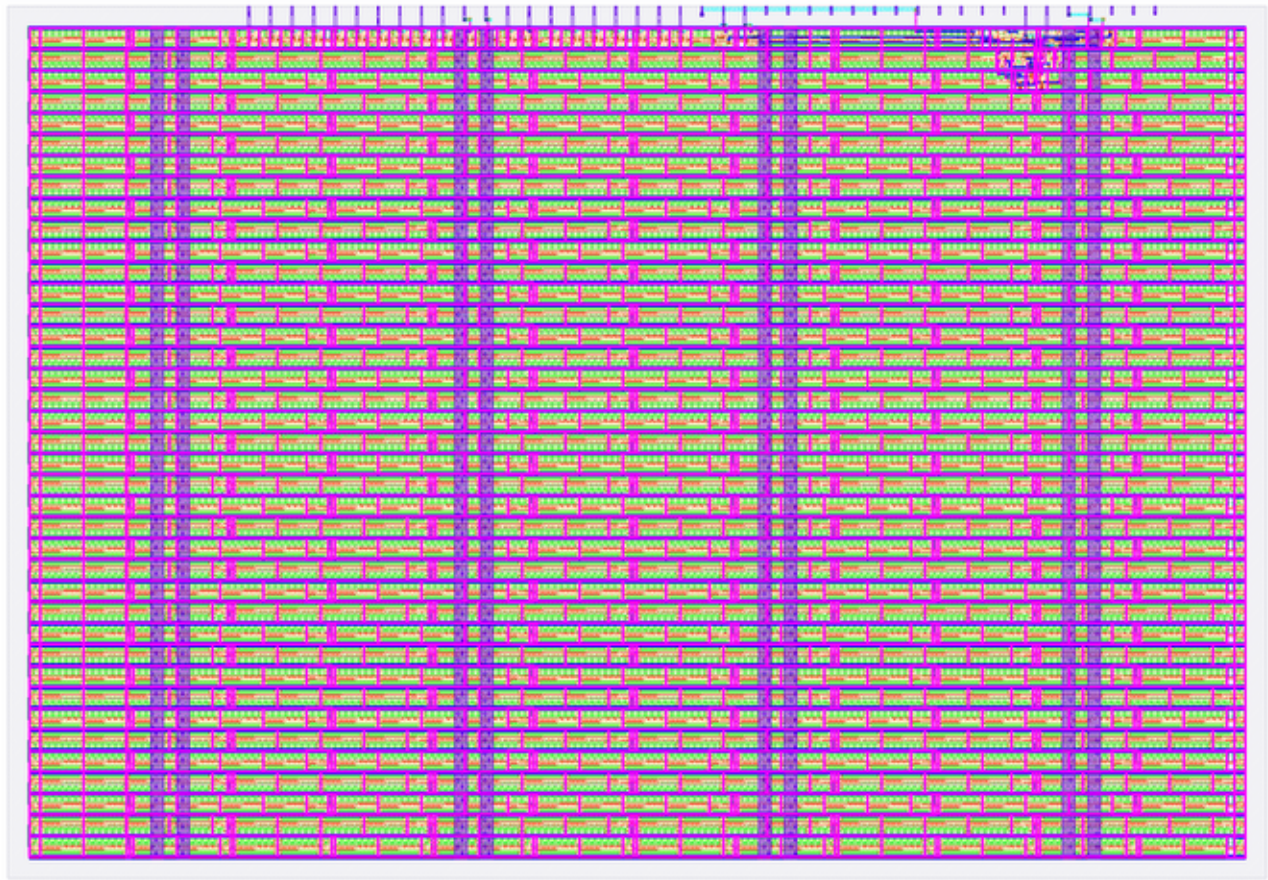
OpenSTA signoff: 9 corners, setup/hold/slew/cap

Corner/Group	Hold Worst Slack	Reg to Reg Paths	Hold TNS	Hold Vio Count	of which reg to reg	Setup Worst Slack	Reg to Reg Paths	Setup TNS	Setup Vio Count	of which reg to reg	Max Cap Violatio…	Max Slew Violati…
Overall	7.9852	N/A	0.0000	0	0	10.3488	N/A	0.0000	0	0	0	0
nom_tt_025C_1v80	8.1066	N/A	0.0000	0	0	11.0026	N/A	0.0000	0	0	0	0
nom_ss_100C_1v60	8.4112	N/A	0.0000	0	0	10.3558	N/A	0.0000	0	0	0	0
nom_ff_n40C_1v95	7.9871	N/A	0.0000	0	0	11.2638	N/A	0.0000	0	0	0	0
min_tt_025C_1v80	8.1039	N/A	0.0000	0	0	11.0060	N/A	0.0000	0	0	0	0
min_ss_100C_1v60	8.4064	N/A	0.0000	0	0	10.3599	N/A	0.0000	0	0	0	0
min_ff_n40C_1v95	7.9852	N/A	0.0000	0	0	11.2672	N/A	0.0000	0	0	0	0
max_tt_025C_1v80	8.1092	N/A	0.0000	0	0	10.9985	N/A	0.0000	0	0	0	0
max_ss_100C_1v60	8.4156	N/A	0.0000	0	0	10.3488	N/A	0.0000	0	0	0	0
max_ff_n40C_1v95	7.9891	N/A	0.0000	0	0	11.2607	N/A	0.0000	0	0	0	0

KEY POINTS

- PVT: ss 100 °C 1.60 V (slow), tt 25 °C 1.80 V, ff −40 °C 1.95 V (fast)
- × min / nom / max wire parasitics = 9 corners
- Setup worst at slow corner (+10.35 ns), hold worst at fast corner (+7.99 ns)
- 20 ns clock is only a reference; IO pads (~33 MHz out) limit real speed
- Power ≈ 5.2 µW (default activity assumption), leakage ≈ 5 nW

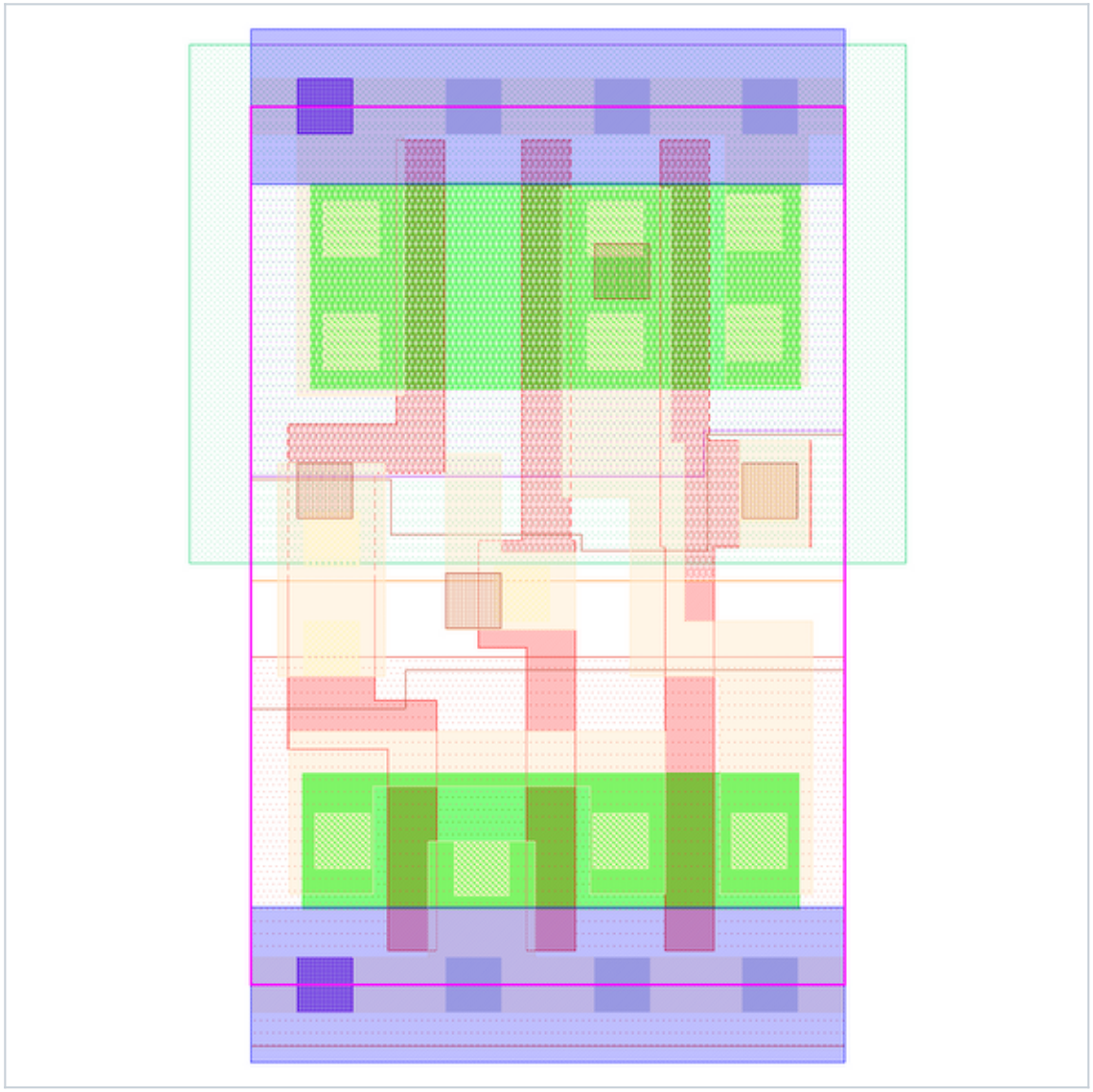
Final GDS (KLayout)



KEY POINTS

- Magic merges OpenROAD placement/routing (DEF) with SkyWater cell layouts (GDS)
- 1,388 filler/decap cells fill empty rows (continuous wells and rails)
- This GDS is what is submitted and fabricated
- PNR view = cell abstracts; GDS = every mask layer (diff, poly, implants …)
- Flow: 80/80 stages in 1 min 13 s

Inside one cell: o21ai_0 (carry gate)

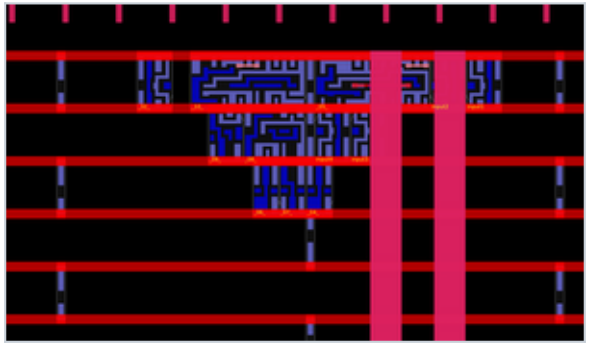
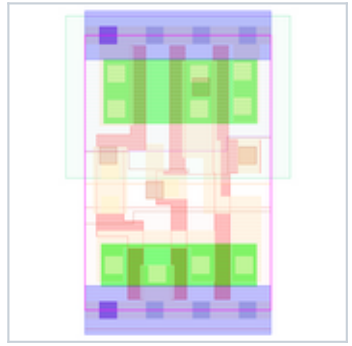


KEY POINTS

- 3 poly gates cross PMOS diffusion (top, in n-well) and NMOS diffusion (bottom)
- 6 transistors, L = 0.15 µm; nfet W 0.42 µm, pfet (hvt) W 0.64 µm
- Matches the cell's SPICE netlist from the PDK
- Pre-designed by SkyWater: the flow places cells, never redraws transistors
- Site 0.46 µm is a placement grid; smallest feature is the 0.15 µm gate

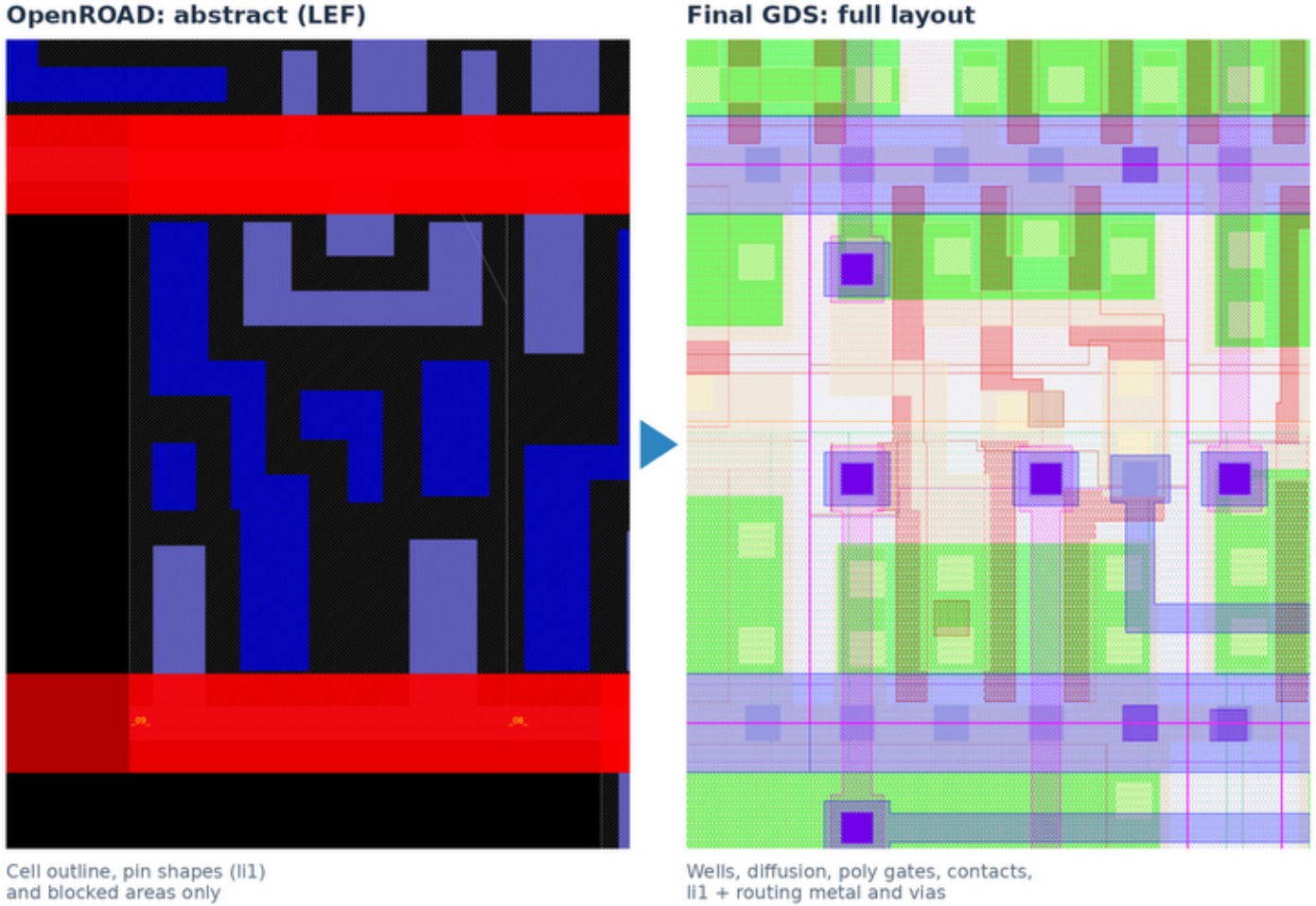
COMPARISON

Digital vs analog layout

	Digital: standard-cell flow (this adder)	Analog: custom layout (by hand)
	 *OpenROAD placement: cells on rows, tap grid, power stripes*	 *One SkyWater cell (o21ai), drawn by hand: transistors placed directly*
Building blocks	Pre-designed standard cells, all one row high	Individual transistors (PDK devices), sized by hand
Placement	Automatic (OpenROAD) on rows and sites	By hand, anywhere; matching and symmetry matter
Rows, endcaps, fill	Required by the standard-cell method	Not used
Well / substrate ties	Tap cells on a regular grid, inserted automatically	Body contacts or guard rings, placed by hand
Power	Generated mesh: met1 rails + met4 stripes	Drawn by hand (Tiny Tapeout: met4 power pins)
Routing	Automatic router	By hand, watching parasitics and coupling
Verification	STA timing, gate-level sim, DRC, LVS	SPICE (post-layout extraction), DRC, LVS
Scales to	Thousands of cells in minutes	Few devices; most control per transistor

Same rules for both: wells and substrate tied, supplies connected, DRC + LVS clean, precheck passed.
Only the method differs: automated (digital) vs by hand (analog).

Abstract view vs full layout: the same cell



KEY POINTS

- Both images: the carry gate _09_ (sky130_fd_sc_hd__o21ai_0, 1.84 × 2.72 µm), same spot in the tile
- Every SkyWater cell comes in several views of ONE cell: Liberty (timing), LEF (abstract), GDS (layout), SPICE
- OpenROAD uses the LEF abstract: outline, pins, blocked areas, enough to place and route, no transistors
- OpenROAD writes a DEF: which cell sits where, plus its routing (already real metal)
- Magic fills each cell in with SkyWater's full layout and keeps the routing → final GDS
- **The full layout (wells, diffusion, poly, contacts, metal) is what is fabricated and checked by DRC/LVS**

DRC and LVS

Magic DRC + Netgen LVS + LibreLane manufacturability summary

```
Magic DRC:
[INFO] COUNT: 0

Netgen LVS:
Netlists match uniquely.
Cell pin lists are equivalent.
Device classes tt_um_tomprusg_adder2 and tt_um_tomprusg_adder2 are equivalent.
Final result: Circuits match uniquely.

LibreLane manufacturability report:
* Antenna
Passed [PASS]

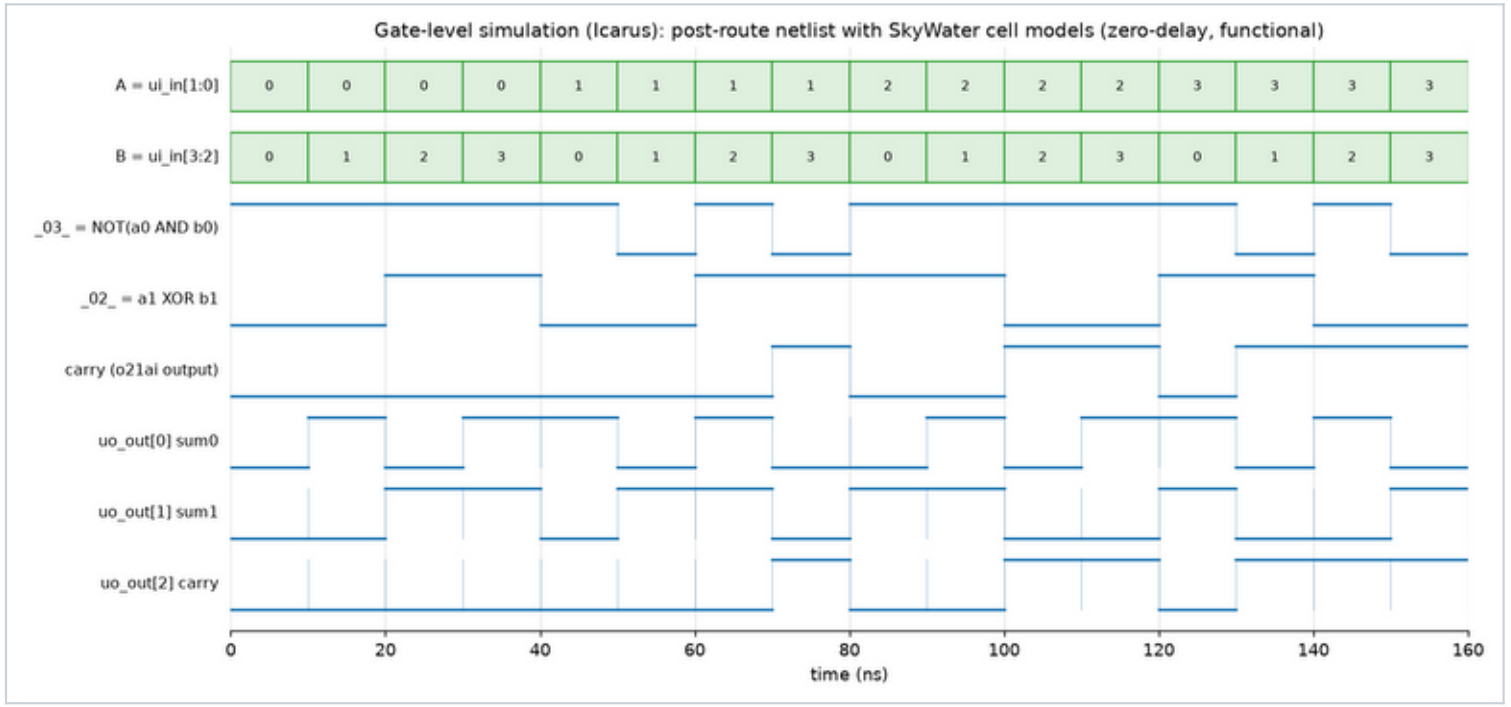
* LVS
Passed [PASS]

* DRC
Passed [PASS]
```

KEY POINTS

- PNR checks (abstract level): legal placement, routing DRC, antenna, timing
- Layout checks (full geometry): Magic DRC = 0 errors
- LVS: transistors extracted from the GDS vs. netlist → circuits match uniquely
- **Antenna, LVS, DRC: passed**

Icarus Verilog: post-route netlist



KEY POINTS

- Same wrapper testbench on the final netlist + SkyWater cell Verilog models
- Includes the added buffers, ties and upsized gates
- **Result: 16/16 correct, PASS**
- Zero-delay (function only): UNIT_DELAY affects only flip-flops, none here
- Timing is covered by STA, not by this simulation

TinyTapeout acceptance checks

TinyTapeout precheck: results

```
Magic DRC                                  PASS
KLayout FEOL                               PASS
KLayout BEOL                               PASS
KLayout offgrid                            PASS
KLayout pin label overlapping drawing      PASS
KLayout zero area                          PASS
KLayout Checks                             PASS
Pin check                                  PASS
Boundary check                             PASS
Power pin check                            PASS
Layer check                                PASS
Cell name check                            PASS
Analog pin check                           PASS
Verilog syntax check                       PASS

INFO: Precheck passed for tt_um_tomprusg_adder2.gds!
```

KEY POINTS

- Magic DRC + KLayout FEOL/BEOL/offgrid DRC (R8)
- Pin check vs. template (R5), boundary 161 × 111.52 (R2)
- Power pins VPWR/VGND on met4 ≥ 1.2 µm (R6); no met5 (R7)
- Cell names, urpm/nwell, analog pins, Verilog syntax
- **15 / 15 checks passed**

Status and next steps

VERIFIED LOCALLY

- ✔ RTL + wrapper simulation 16/16
- ✔ Synthesis: 29 cells, 126 µm²
- ✔ Place & route: 0 violations
- ✔ Timing: +10.35 ns setup, +7.99 ns hold, 9 corners
- ✔ DRC 0, LVS match, antenna 0
- ✔ Gate-level simulation 16/16
- ✔ TinyTapeout precheck 15/15

REMAINING

- Adapt official cocotb tests (test/) to the adder
- Write the datasheet (info.md)
- Repository hygiene (.venv ignore, line endings)
- Push to GitHub: official build with LibreLane 3.0.14 + PDK 8afc8346 = final acceptance

Sources (1/2): TinyTapeout constraint files

Bold = used by the tools in this flow Grey = reference / background reading *Links are clickable · pinned to exact versions*

TinyTapeout support tools · commit b7b993a

tile_sizes.yaml: die size per tile	`tt-support-tools/tech/sky130A/tile_sizes.yaml`
tech.py: SKY130 settings (top metal met4)	`tt-support-tools/tech.py`
project.py: port check, config generation	`tt-support-tools/project.py`
tt_block_1x1_pg.def: 1x1 pin template	`tt-support-tools/tech/sky130A/def/tt_block_1x1_pg.def`
tt_tool.py: command-line entry point	`tt-support-tools/tt_tool.py`
precheck/tech_data.py: layer and power-pin rules	`tt-support-tools/precheck/tech_data.py`
precheck/precheck.py: acceptance checks	`tt-support-tools/precheck/precheck.py`

TinyTapeout digital project template · commit a42e666

src/config.json: template flow settings	`ttsky-verilog-template/src/config.json`
info.yaml: template metadata	`ttsky-verilog-template/info.yaml`
.github/workflows/gds.yaml: official CI	`ttsky-verilog-template/.github/workflows/gds.yaml`
src/project.v: example wrapper	`ttsky-verilog-template/src/project.v`

Sources (2/2): build action, LibreLane, PDK, documentation

● **Bold = used by the tools in this flow** Grey = reference / background reading *Links are clickable · pinned to exact versions*

TinyTapeout build action · tag ttsky26d (LibreLane 3.0.14)

action.yml: GDS build used by CI	`tt-gds-action/action.yml @ ttsky26d`
precheck/action.yml: CI precheck	`tt-gds-action/precheck/action.yml @ ttsky26d`

LibreLane

● **base.sdc: timing constraints script**	**`librelane/…/base.sdc @ 3.1.0.dev3 (local run)`**
pdk_hashes.yaml: PDK revision expected by CI	`librelane/…/pdk_hashes.yaml @ 3.0.14 (CI)`

SkyWater PDK LibreLane config · open_pdks commit 1689ac3f

● **config.tcl: tap distance, corner, RT_MAX_LAYER**	**`open_pdks/sky130/librelane/config.tcl`**
● **sky130_fd_sc_hd/: site, taps, driver, load, slew**	**`open_pdks/sky130/librelane/sky130_fd_sc_hd/`**

Documentation

TinyTapeout specifications (tiles, pins)	`tinytapeout.com/specs`
TinyTapeout GPIO / pad limits	`tinytapeout.com/specs/gpio`
TinyTapeout local hardening guide	`tinytapeout.com/guides/local-hardening`
LibreLane configuration variables	`librelane.readthedocs.io/…/configuration.html`
Chip multiplexer architecture (slides 2–4 images)	`tt-multiplexer/docs/INFO.md`
ETR demo board (slide 3 photo)	`tinytapeout.com/specs/pcb-etr`
Real die photos: TT07 interactive viewer	`tinytapeout.com/decap/tt07`